AgentNexus: A Service-Boundary-Aware Coordination Architecture for Heterogeneous LLM Code Agents

dugubuyan Independent Researcher GitHub: github.com/dugubuyan · X: @dugubuyan

April 2026

Keywords: multi-agent systems, LLM agents, software engineering, service-oriented architecture, publish-subscribe, Model Context Protocol, lifecycle management

---

Abstract

Existing multi-agent software development frameworks such as ChatDev and MetaGPT organize agents around *roles* (product manager, developer, tester) within a single simulated organization. While effective for monolithic tasks, this role-playing paradigm breaks down in real-world polyglot systems where multiple independently-deployed services—each maintained by its own LLM agent—must coordinate across service boundaries. We present **AgentNexus**, a document exchange center that coordinates heterogeneous code agents at the *service* granularity rather than the role granularity. AgentNexus introduces three key ideas: (1) a versioned Markdown document store with publish-subscribe notification, enabling agents to detect and respond to cross-service changes; (2) an explicit lifecycle stage model that tracks each service's development phase as a first-class entity, replacing ad-hoc role-playing with structured state transitions; and (3) a diff-aware update protocol that delivers structured change summaries alongside full document context, allowing downstream agents to perform targeted code modifications. We describe the architecture, implementation, and an initial deployment coordinating a backend search service and its frontend management console. Our results suggest that grounding multi-agent coordination in service-level document exchange—rather than simulated organizational roles—better reflects the structure of real software systems and reduces coordination overhead.

---

1. Introduction

The past two years have seen rapid progress in LLM-based multi-agent systems for software engineering. Pioneering frameworks such as ChatDev [Qian et al., 2024] and MetaGPT [Hong et al., 2024] demonstrated that a collection of role-playing agents—simulating product managers, architects, developers, and testers—can autonomously produce working software from natural-language requirements. These systems adopt a *role-centric* coordination model: agents are assigned human organizational roles, and coordination happens through simulated meetings, code reviews, and document handoffs within a single shared context.

This role-centric model has a fundamental mismatch with real-world software development at scale. Production systems are not monolithic; they are composed of multiple independently-deployed services, each with its own codebase, technology stack, and development team. When a backend API changes, the frontend must adapt. When a shared configuration changes, every dependent service must update. These cross-service dependencies are not captured by role assignments—they are captured by *service boundaries and the documents that cross them*.

We observe that the core coordination problem in multi-service development is not "which role should handle this task" but rather "which service needs to know about this change, and what exactly changed." This reframing leads us to a fundamentally different architecture.

We present **AgentNexus**, a document exchange center that acts as the coordination substrate for a collection of LLM code agents, each responsible for a distinct service. AgentNexus makes three contributions:

1. **Service-granular coordination**: Each agent is registered as a *sub-project* with its own document namespace. Documents (requirements, design, API specs, configuration) are versioned and stored per service. Agents subscribe to documents from other services they depend on.

2. **Lifecycle stage as a first-class entity**: Rather than simulating organizational roles, AgentNexus tracks each service's development lifecycle stage (design → development → testing → deployment → upgrade) as a persistent, queryable attribute. Stage transitions trigger milestone snapshots and cross-service notifications, grounding coordination in the actual state of the system.

3. **Diff-aware update protocol**: When a subscribed document changes, the `get_my_updates_with_context` API delivers both a structured diff (unified diff format) and the full latest document in a single call. This allows downstream agents to perform targeted, context-aware code modifications rather than full re-reads.

---

2. Background and Related Work

2.1 Role-Playing Multi-Agent Frameworks

ChatDev [Qian et al., 2024] organizes agents as CEO, CTO, programmer, and tester, coordinating through a "chat chain" of sequential dialogues. MetaGPT [Hong et al., 2024] introduces Standardized Operating Procedures (SOPs) and assigns agents roles such as product manager and QA engineer, producing structured artifacts in a waterfall-style pipeline. ALMAS [2025] extends this to agile workflows with sprint planning and code review agents.

These frameworks share a common assumption: all agents operate within a single simulated organization on a single codebase. Coordination is achieved through shared context and role-based task delegation.

2.2 Limitations of the Role-Centric Model

He et al. [2024] identify several open challenges in LLM-based multi-agent software engineering, including context window limitations, agent misalignment, and the difficulty of managing long-horizon tasks. E2EDev [2025] benchmarks show that multi-agent frameworks do not consistently outperform single-agent approaches, partly due to coordination overhead.

A deeper limitation, less discussed in the literature, is *organizational boundary mismatch*. Real software systems are not single organizations—they are ecosystems of services. The role-playing metaphor forces a flat organizational structure onto what is inherently a distributed, service-oriented architecture. When a backend developer agent and a frontend developer agent are both "developers" in the same simulated company, there is no natural mechanism to enforce service boundaries, version contracts, or change propagation.

2.3 Publish-Subscribe for Agent Coordination

The publish-subscribe pattern [Eugster et al., 2003] is well-established in distributed systems for decoupling producers from consumers. Recent work on agent interoperability protocols, including MCP [Anthropic, 2024] and A2A [Google, 2025], has begun to apply similar ideas to agent communication. AgentNexus extends this pattern specifically to *document-level* coordination in software development, where the "messages" are versioned Markdown documents representing service contracts.

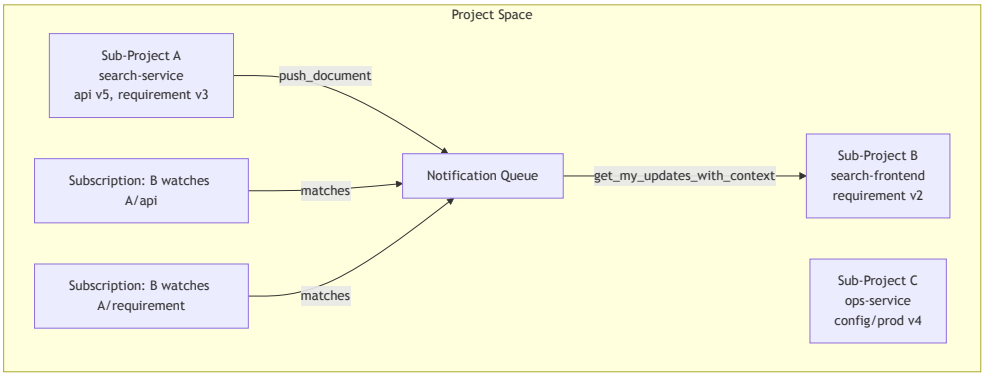
---

3. Architecture

3.1 Core Abstractions

AgentNexus organizes the world around four abstractions:

Figure 1 shows the overall system structure. A Project Space contains multiple Sub-Projects (services), each owning a set of versioned Documents. Subscriptions connect sub-projects across service boundaries, and the notification queue delivers change events to subscribers.



diagram

Figure 1: AgentNexus data model. Sub-projects own documents; subscriptions define cross-service dependencies; notifications propagate changes to subscribers.

Project Space: The top-level isolation unit, corresponding to a large project or product. All sub-projects, documents, and subscriptions belong to a space.

Sub-Project: A registered service or component, identified by a UUID `project_id`. Each sub-project has a name, type (development, testing, ops, infra), and lifecycle stage.

Document: A versioned Markdown artifact belonging to a sub-project. Documents are typed: `requirement`, `design`, `api`, `config`, or `task`. Each push creates a new version; content is deduplicated by SHA-256 hash.

Subscription: A rule declaring that sub-project A should be notified when a specific document (or document type) from sub-project B changes.

3.2 Lifecycle Stage Model

Each sub-project carries a `stage` attribute drawn from a fixed vocabulary: `design`, `development`, `testing`, `deployment`, `upgrade`. Stage transitions are explicit operations that:

1. Update the sub-project's stage and record the transition timestamp.

2. Automatically create *milestone snapshots*—immutable copies of all published documents at the moment of transition.
3. Generate stage-switch tasks for affected sub-projects.

This model differs fundamentally from role-playing frameworks. Rather than assigning an agent the *role* of "tester," AgentNexus records that a service is *in the testing stage*. The distinction matters: a service can be in the testing stage while its dependent frontend is still in development. The stage is a property of the service, not of an agent persona.

3.3 Diff-Aware Update Protocol

When a subscribed document is updated, AgentNexus generates a notification containing the new version number. When an agent calls `get_my_updates_with_context`, the system returns, for each unread notification:

- `diff`: A unified diff between the previous and current version, computed server-side using Python's `difflib`.
- `latest_content`: The full text of the current version.
- `doc_type`: The document type, enabling agents to route updates to appropriate handlers.

This design reflects a key insight: agents need both *what changed* (to perform targeted modifications) and *the full current state* (to maintain correct context). Providing only the diff risks missing context; providing only the full document makes it difficult to identify the locus of change.

3.4 MCP Interface

AgentNexus exposes its functionality as a Model Context Protocol server running in streamable-HTTP mode, allowing multiple agents to connect simultaneously. The tool set is divided into:

Agent tools: `push_document`, `get_document`, `get_my_updates_with_context`, `ack_update`, `get_my_tasks`, `get_config`

Admin tools: `create_space`, `register_project`, `list_projects`, `add_subscription`, `publish_draft`, `generate_steering_file`, `get_project_id_by_name`

3.5 Steering File Integration

Each sub-project's IDE agent is configured with a *steering file*—a Markdown document loaded into the agent's context at startup. The steering file instructs the agent to:

1. Resolve its `project_id` by name at startup via `get_project_id_by_name`.

2. Call `get_my_updates_with_context` to check for pending document changes.
3. Apply changes based on diff and full context, then acknowledge via `ack_update`.
4. Push updated documents after significant code changes via `push_document`.

This creates a self-contained coordination loop that requires no human intervention once configured.

---

4. Comparison with Role-Centric Frameworks

Dimension	Role-Centric (ChatDev, MetaGPT)	AgentNexus
Coordination unit	Agent role (developer, tester)	Service (sub-project)
Lifecycle tracking	Implicit in workflow phase	Explicit stage attribute per service
Change propagation	Shared context / sequential handoff	Pub-sub notification with versioned diff
Service boundaries	Not enforced	First-class: each service has its own document namespace
Multi-codebase support	Single codebase assumed	Native: each sub-project is an independent repository
Human oversight	Checkpoint prompts	Admin tools + milestone snapshots
Context management	Full conversation history	Targeted diff + full document on demand

The key architectural difference is that AgentNexus treats the *service* as the unit of coordination, not the *agent role*. This allows agents to be heterogeneous—different LLMs, different IDEs, different programming

languages—as long as they speak the MCP protocol and follow the document exchange contract.

---

5. Implementation

AgentNexus is implemented in Python using:

- **FastMCP** (mcp[cli] >= 1.0) for the MCP server layer
- **SQLAlchemy** + **SQLite** for document storage (with a migration path to PostgreSQL)
- **Alembic** for schema migrations
- **watchdog** for filesystem-based document ingestion
- **difflib** for server-side diff computation

The system runs as a single persistent process, exposing the MCP endpoint at `http://0.0.0.0:10086/mcp`. The FileWatcherService monitors a `/docs/` directory, automatically ingesting Markdown files written by agents as draft documents.

The full implementation includes 191 unit and property-based tests using the Hypothesis framework. The source code is available at https://github.com/dugubuyan/agent-nexus.

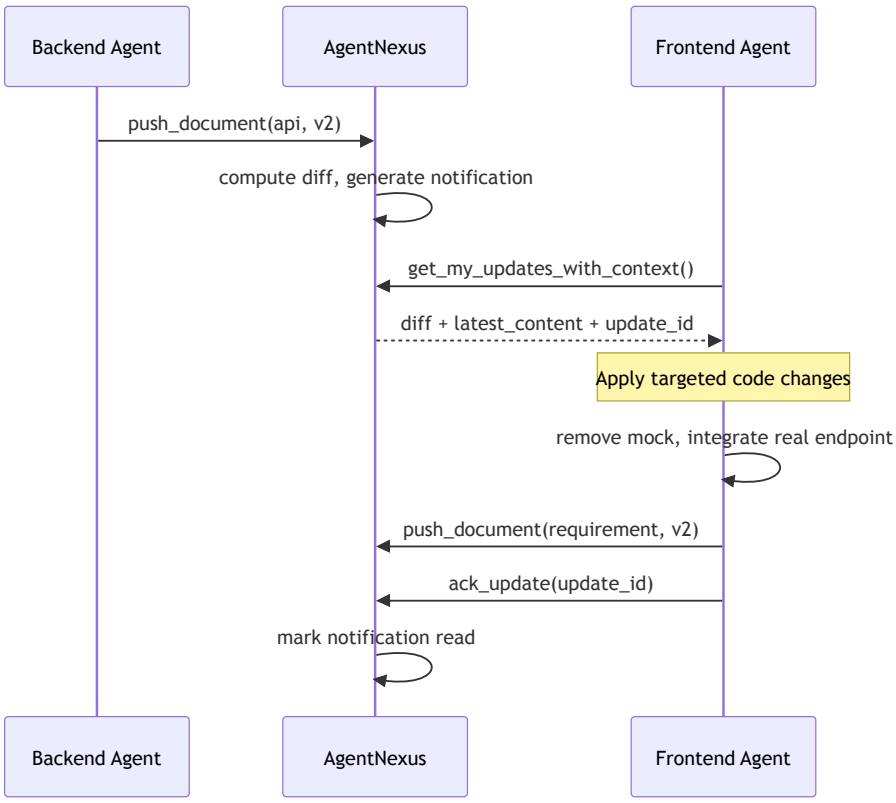
---

6. Initial Deployment

We deployed AgentNexus to coordinate two services in a financial information retrieval system:

- **search-service**: A Python/FastAPI backend providing full-text search over Elasticsearch, with admin endpoints for document review, pipeline monitoring, and sensitive word management.
- **search-admin-frontend**: A React/Ant Design management console consuming the search-service admin APIs.

The frontend sub-project subscribes to the search-service's `api` and `requirement` documents. **Figure 2** illustrates the end-to-end coordination flow when the backend implements a new endpoint.



diagram

Figure 2: End-to-end coordination flow. The backend agent pushes an updated API document; AgentNexus delivers a diff-aware notification; the frontend agent makes targeted code changes and acknowledges.

When the backend team implements a new endpoint (`PUT /admin/docs/{doc_id}` for in-place document editing), the workflow proceeds as follows:

1. The backend agent updates `search-service/api` via `push_document`.
2. AgentNexus generates a notification for `search-admin-frontend`.
3. The frontend agent calls `get_my_updates_with_context`, receiving the diff showing the new endpoint and the full updated API document.
4. The frontend agent removes the mock implementation and integrates the real endpoint.
5. The frontend agent updates its own `requirement` document to remove the "backend not yet implemented" annotation.
6. The frontend agent calls `ack_update` to mark the notification as read.

This end-to-end flow requires no human coordination beyond the initial subscription configuration.

---

7. Discussion

7.1 Service Boundary as Coordination Primitive

The central claim of this paper is that *service boundaries*, not *agent roles*, are the appropriate primitive for coordinating LLM agents in real software development. This claim is grounded in the observation that real software systems are already organized around service boundaries—microservices, APIs, configuration contracts—and that the coordination problems that arise in practice (interface drift, configuration mismatch, undocumented changes) are fundamentally cross-service problems.

Role-playing frameworks address a different problem: how to decompose a single development task among multiple agents. AgentNexus addresses the complementary problem: how to keep multiple independently-developed services aligned over time.

7.2 Stage as System State

The lifecycle stage model in AgentNexus reflects a view of software development as a *stateful process* rather than a *sequence of role activations*. When a service transitions from development to testing, this is a meaningful event that should trigger downstream actions (milestone snapshots, cross-service notifications, task generation). Encoding this as a first-class system attribute—rather than as a prompt instruction to an agent playing the role of "scrum master"—makes the state observable, queryable, and auditable.

7.3 Limitations and Future Work

The current implementation has several limitations. First, the diff-based change detection is purely textual; semantic understanding of what a change *means* for dependent services requires LLM reasoning, which AgentNexus delegates to the consuming agent. Future work could integrate an LLM-based impact analyzer to generate natural-language change summaries alongside the raw diff.

Second, the subscription configuration is currently manual. Future work will automate subscription inference by analyzing design documents to identify cross-service dependencies.

Third, the system currently uses SQLite, which limits concurrent write throughput. Migration to PostgreSQL with row-level locking will be necessary for larger deployments.

---

8. Conclusion

We have presented AgentNexus, a document exchange architecture that coordinates LLM code agents at the service granularity. By treating services—not roles—as the unit of coordination, and by making lifecycle stage a first-class system attribute, AgentNexus provides a coordination substrate that better matches the structure of real software systems than existing role-playing frameworks. The diff-aware update protocol enables targeted, context-aware code modifications across service boundaries. We believe this service-boundary-aware approach represents a promising direction for scaling LLM-based software development to real-world polyglot systems.

References

- Qian, C. et al. (2024). ChatDev: Communicative Agents for Software Development. *ACL 2024*.
- Hong, S. et al. (2024). MetaGPT: Meta Programming for A Multi-Agent Collaborative Framework. *ICLR 2024*.
- He, J., Treude, C., Lo, D. (2024). LLM-Based Multi-Agent Systems for Software Engineering: Literature Review, Vision and the Road Ahead. *arXiv:2404.04834*.
- ALMAS (2025). An Autonomous LLM-based Multi-Agent Software Engineering Framework. *arXiv:2510.03463*.
- E2EDev (2025). Benchmarking Large Language Models in End-to-End Software Development Task. *arXiv:2510.14509*.
- Anthropic (2024). Model Context Protocol. *anthropic.com/news/model-context-protocol*.
- Eugster, P. et al. (2003). The Many Faces of Publish/Subscribe. *ACM Computing Surveys*.
- RTADev (2025). Intention Aligned Multi-Agent Framework for Software Development. *ACL 2025 Findings*.